

Introduction to Blockchain

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Why we are here?



Syllabus

- Launching coins on Pump.fun
- NFTs
- Trade signals v TG
- Crypto trading



Syllabus

- Launching coins on Pump.fun
- NFTs
- Trade signals v TG
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True syllabus

- Basic cryptography
- Bitcoin
- Consensus algorithms
- Ethereum Virtual Machine (EVM)
- Smart contracts
- DeFi patterns
- DEX protocols
- Lending protocols
- Staking protocols
- Bridges
- Layer 2 solutions
- ZK

Evaluation

$$O_{total} = [0.7 \cdot O_{nakop} + 0.3 \cdot O_{exam}]$$

$$O_{nakop} = 0.6 \cdot O_{CTF} + 0.3 \cdot O_{projects} + 0.1 \cdot O_{quiz}$$

$$[O_{nakop}] \geq 8 \Rightarrow O_{total} = [O_{nakop}]$$

CTF

- CTF (Capture The Flag) challenges with blockchain interaction.
- Start after ~4th lecture.
- Soft and hard deadlines on tasks.

Projects

- The goal: develop the ability to explore a topic in greater depth.
- Can be research or software project.
- A list of topics will be provided, but students can propose their own.

Quiz

- The goal: lectures marketing.
- After each lecture, a short quiz of 5 questions will be given on the covered material.

Exam

- The exam will be conducted in an oral format.
- The list of questions will be provided in advance.

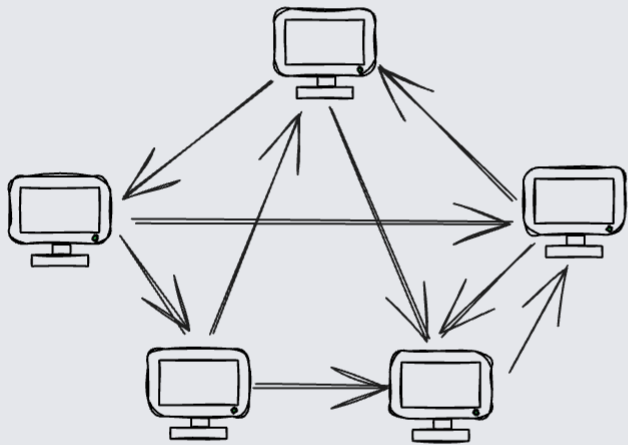
Questions?

Agenda today

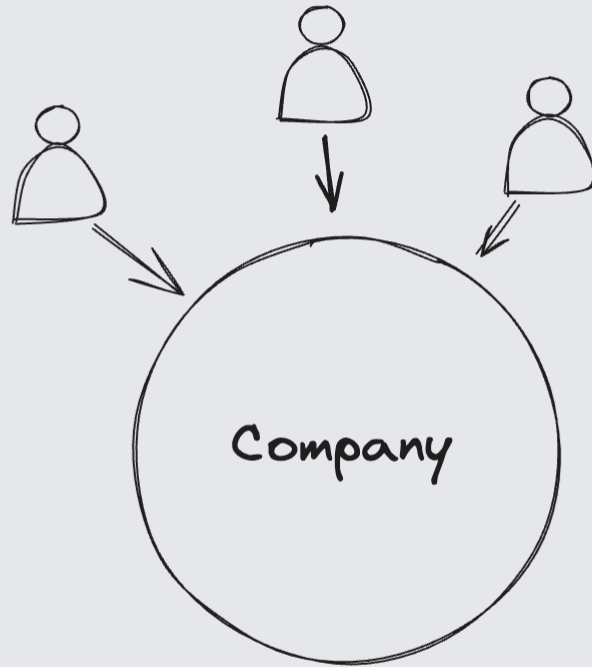
- History
- What is a blockchain?
- Cryptography background
- Crypto wallets

History

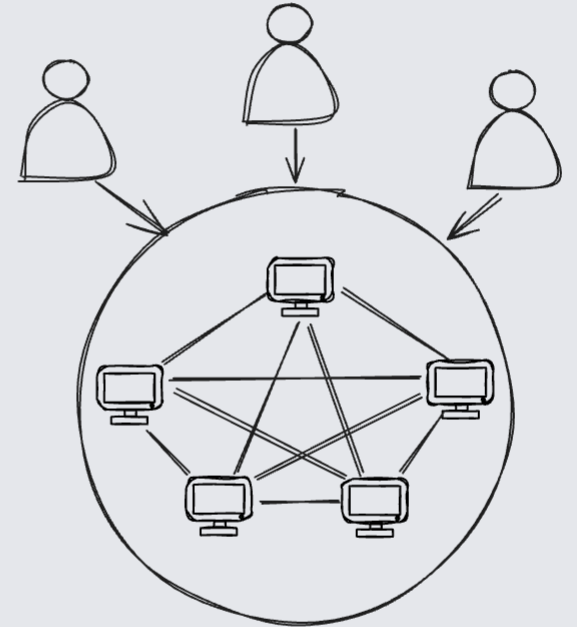
Three eras of networks



Protocol Networks

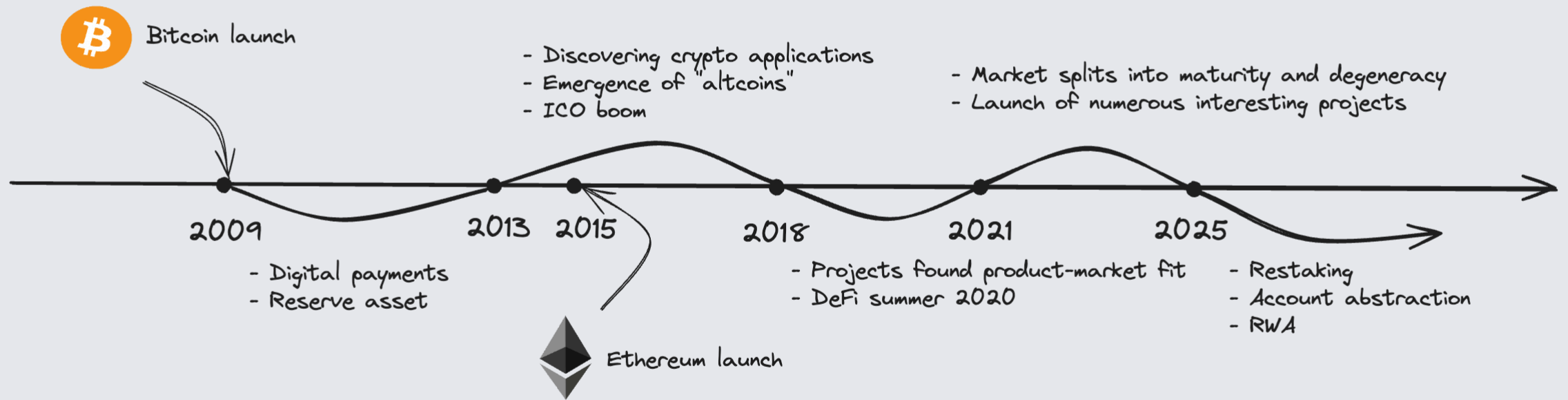


Corporate Networks

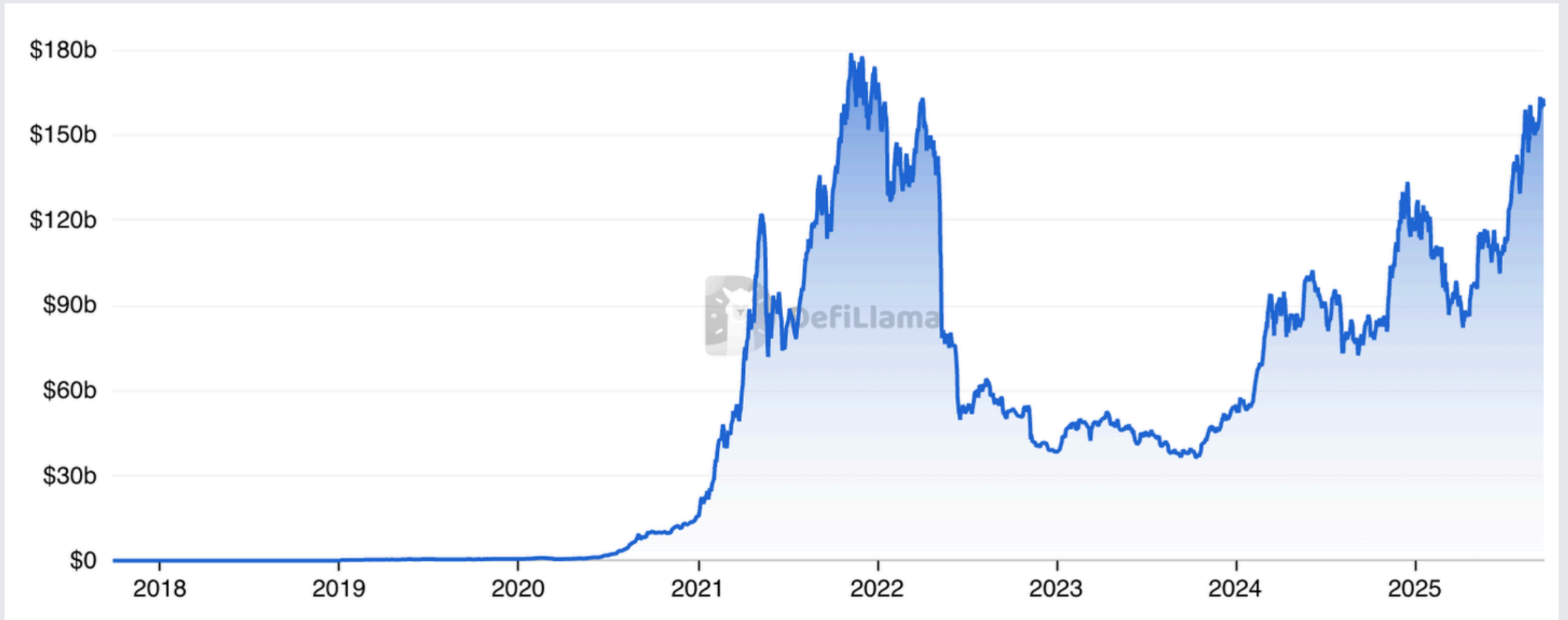


Blockchain Networks

Timeline



TVL (Total Value Locked)



**What is a
blockchain?**



Why is a blockchain?

- A system for coordinating between many parties when no single trusted party exists.
- If trusted party exists $\Rightarrow$ No need for blockchain
- Often used in financial systems

Decentralized ledger

100\$



Alice

Person	Balance
Alice	1000\$
Bob	10\$
...	...



Bob

Decentralized ledger



Alice

Person	Balance
Alice	900\$
Bob	110\$
...	...

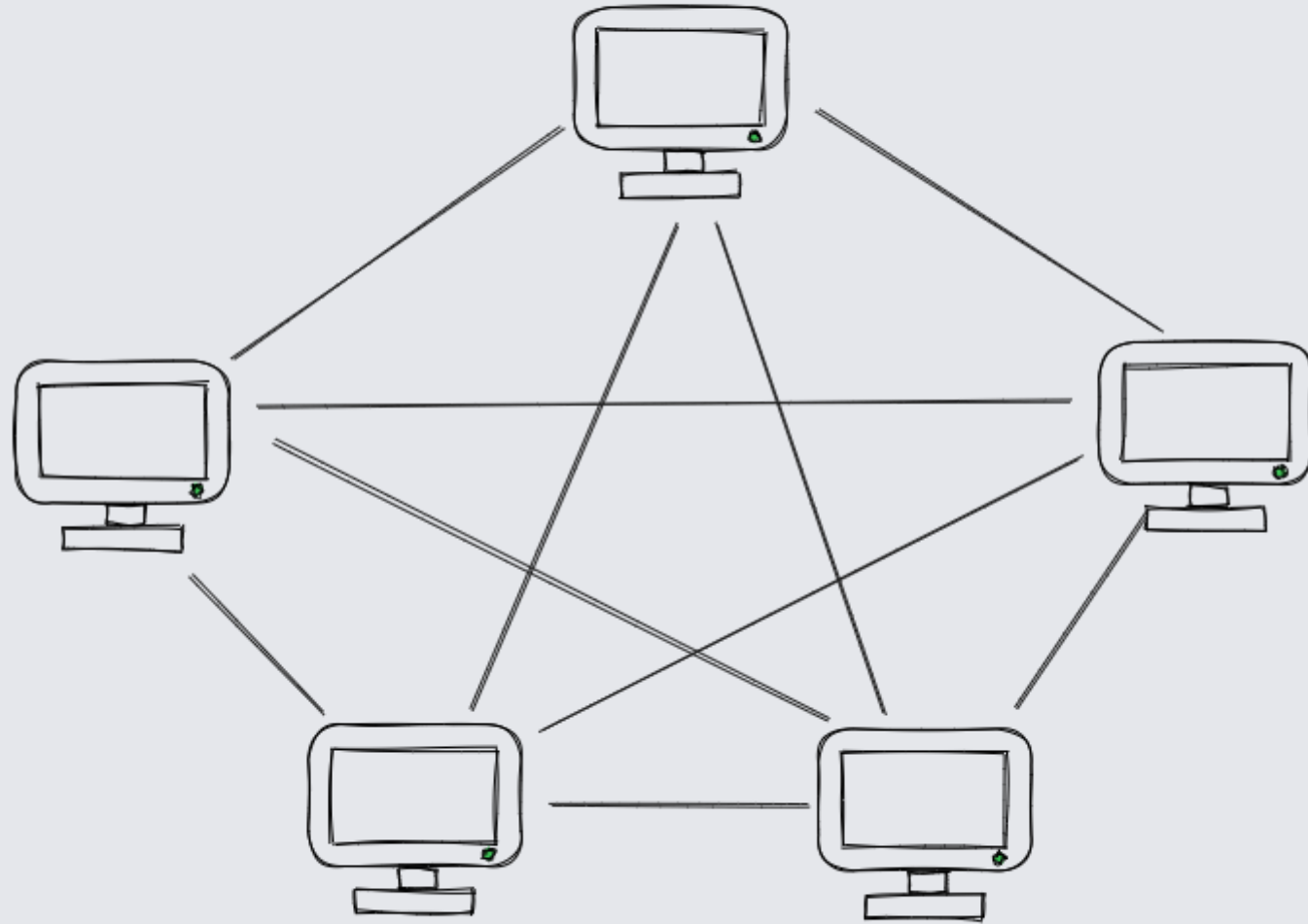


Bob

Requirements for the system

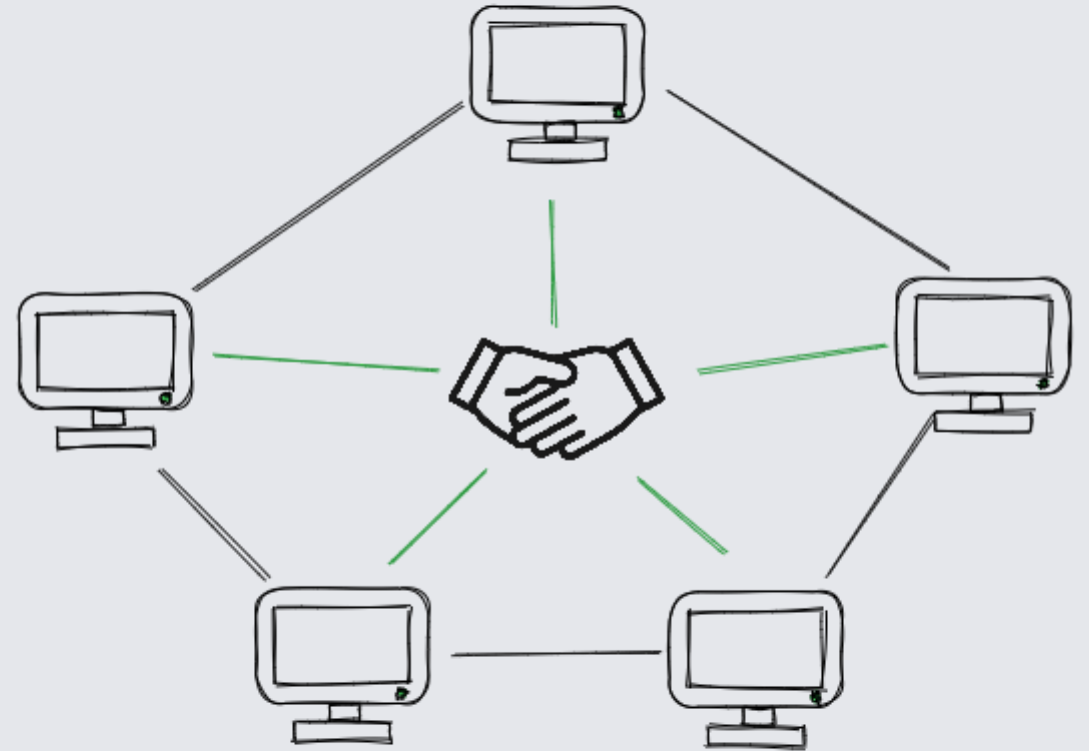
- Broadcast signed messages (transactions).
- Performs state transitions.
- Doesn't allow overspending.
- Decentralized.

Peer-to-Peer (P2P) network



Consensus mechanism

- The term refers to the mechanics, that allow a network of nodes to agree on the state of a blockchain.



Consensus types

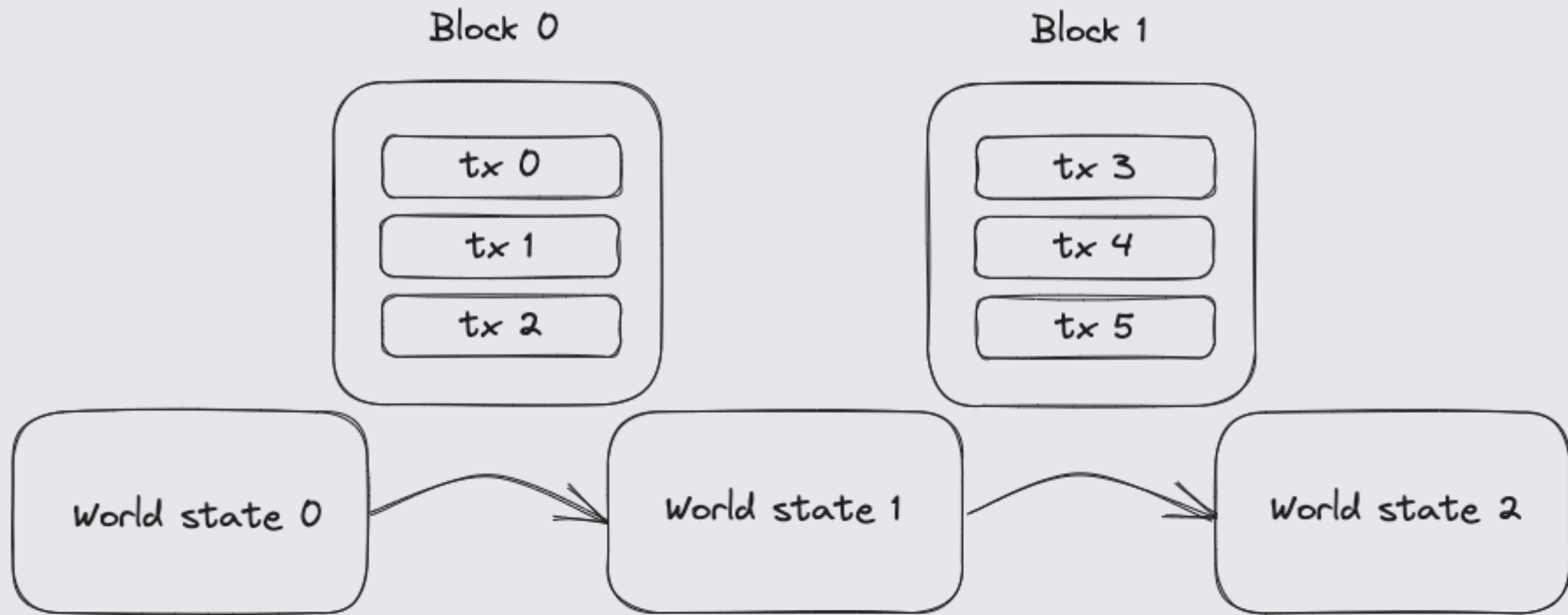
Proof of Work (PoW)

- Block creators are called miners.
- Miners run software that solves complex mathematical problems to validate and add new transactions.
- Mining consumes significant electrical power.

Proof of Stake (PoS)

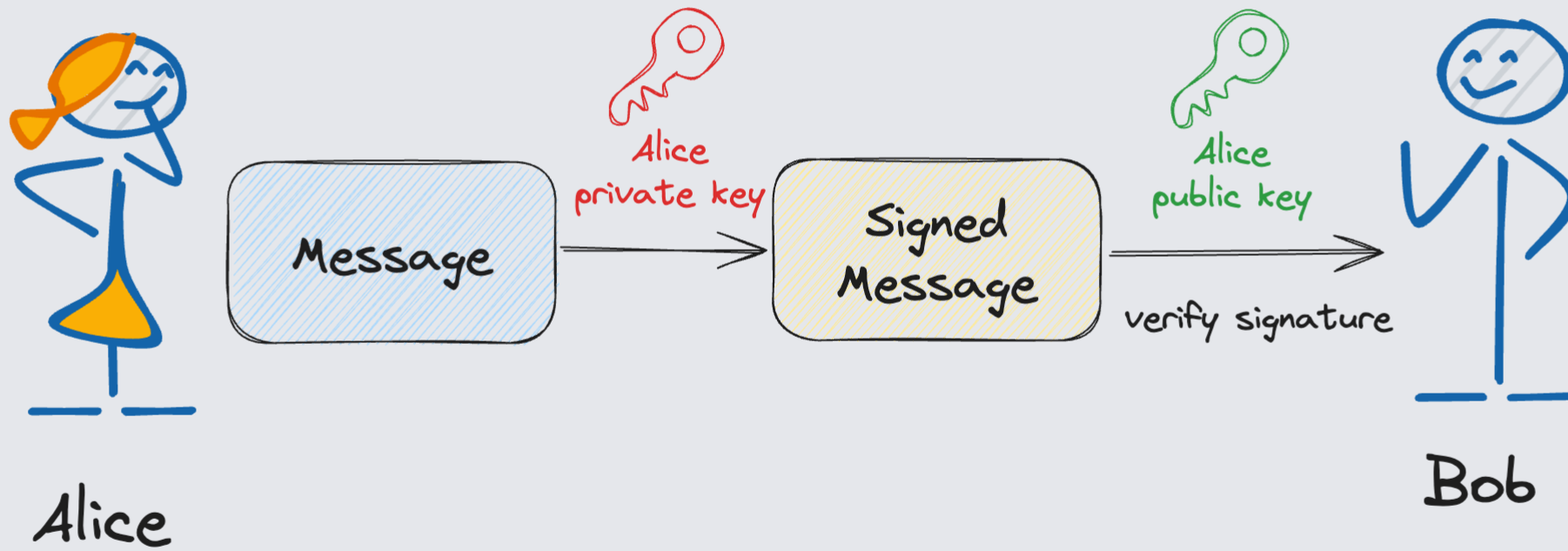
- Block creators are called validators.
- Validators lock up a certain amount of cryptocurrency as collateral to validate and produce new blocks.
- Malicious validators will be losing their staked assets if they attempt to validate fraudulent transactions.

Chain of states



Cryptography background

Digital signature

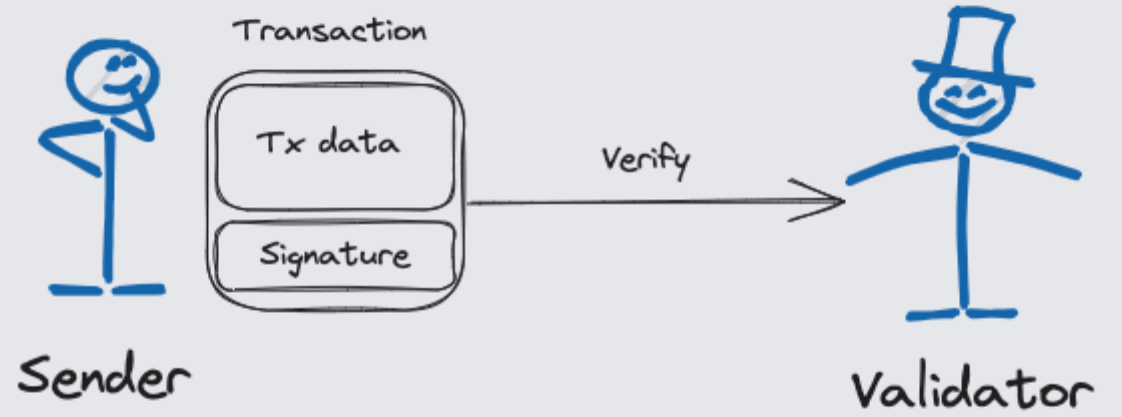


Digital signature: algorithms

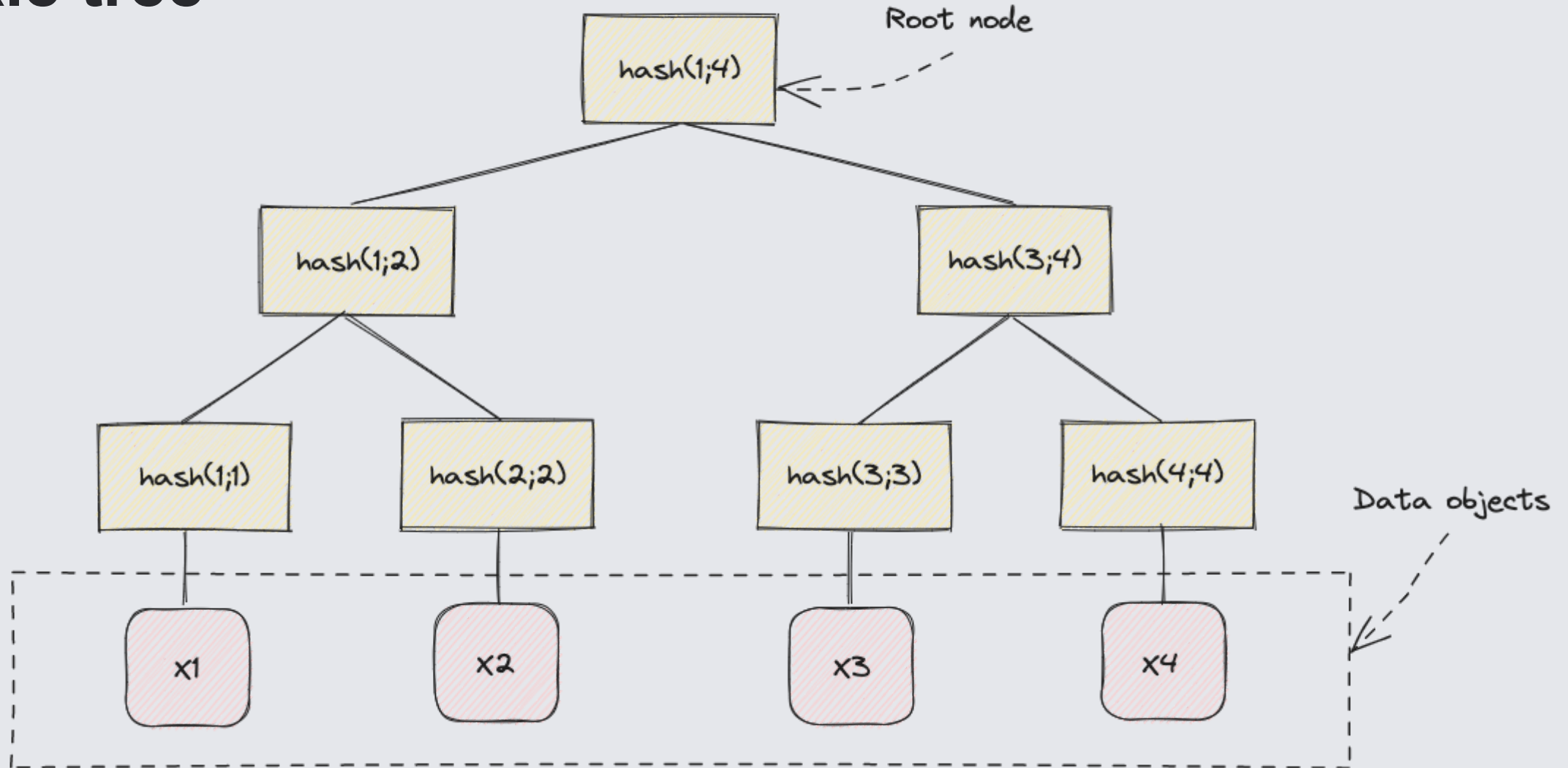
- `gen()` – generate a key pair `(public_key, private_key)`.
- `sign(data, private_key)` – signs data.
- `verify(signature, public_key)` – verifies signature or extract the data.

Signatures on blockchain

- Signing user transactions.
- Signing blocks (for validators).
- Other applications, such as voting mechanics.



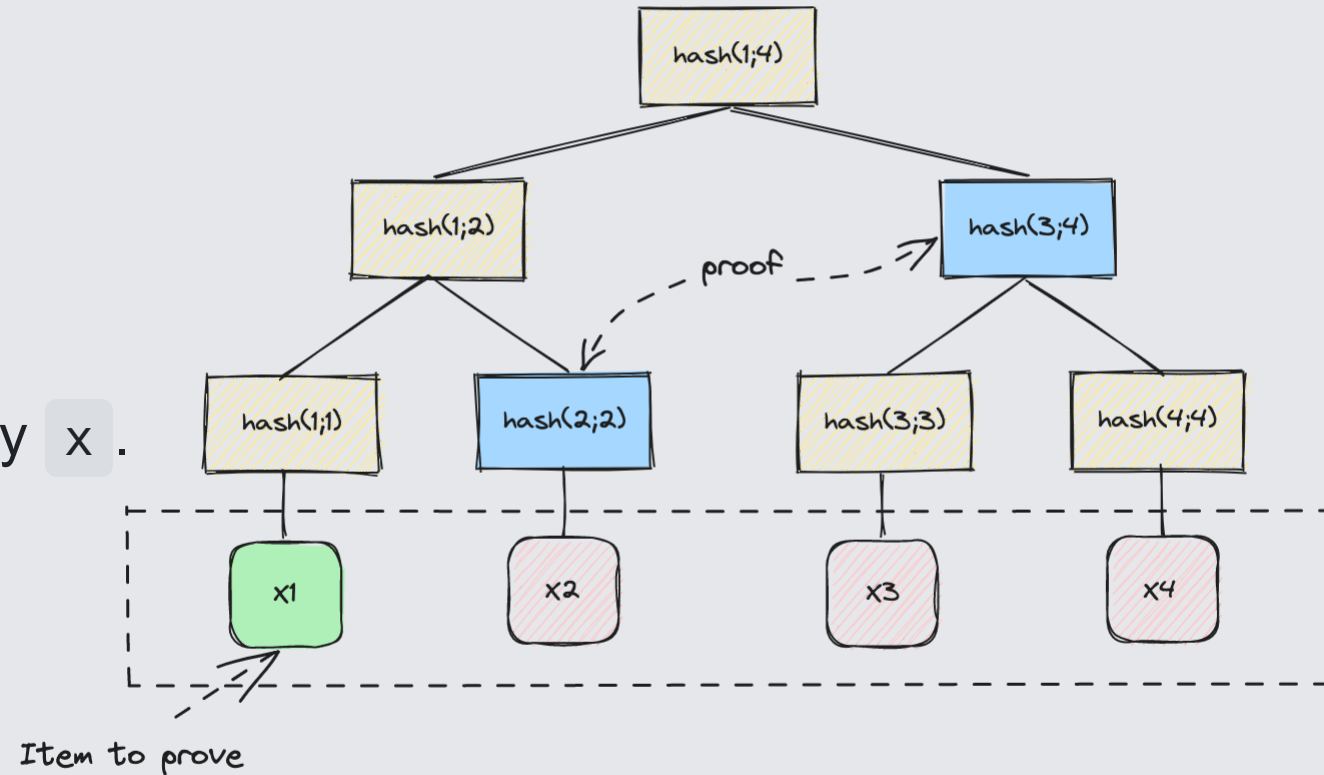
Merkle tree



Merkle tree

Goal

- get a short representation to the array x .
- later prove that element i is $x[i]$.



Merkle tree on blockchain

- Each block contains only merkle root of transactions.
- Later we can prove that a transaction is on the blockchain.

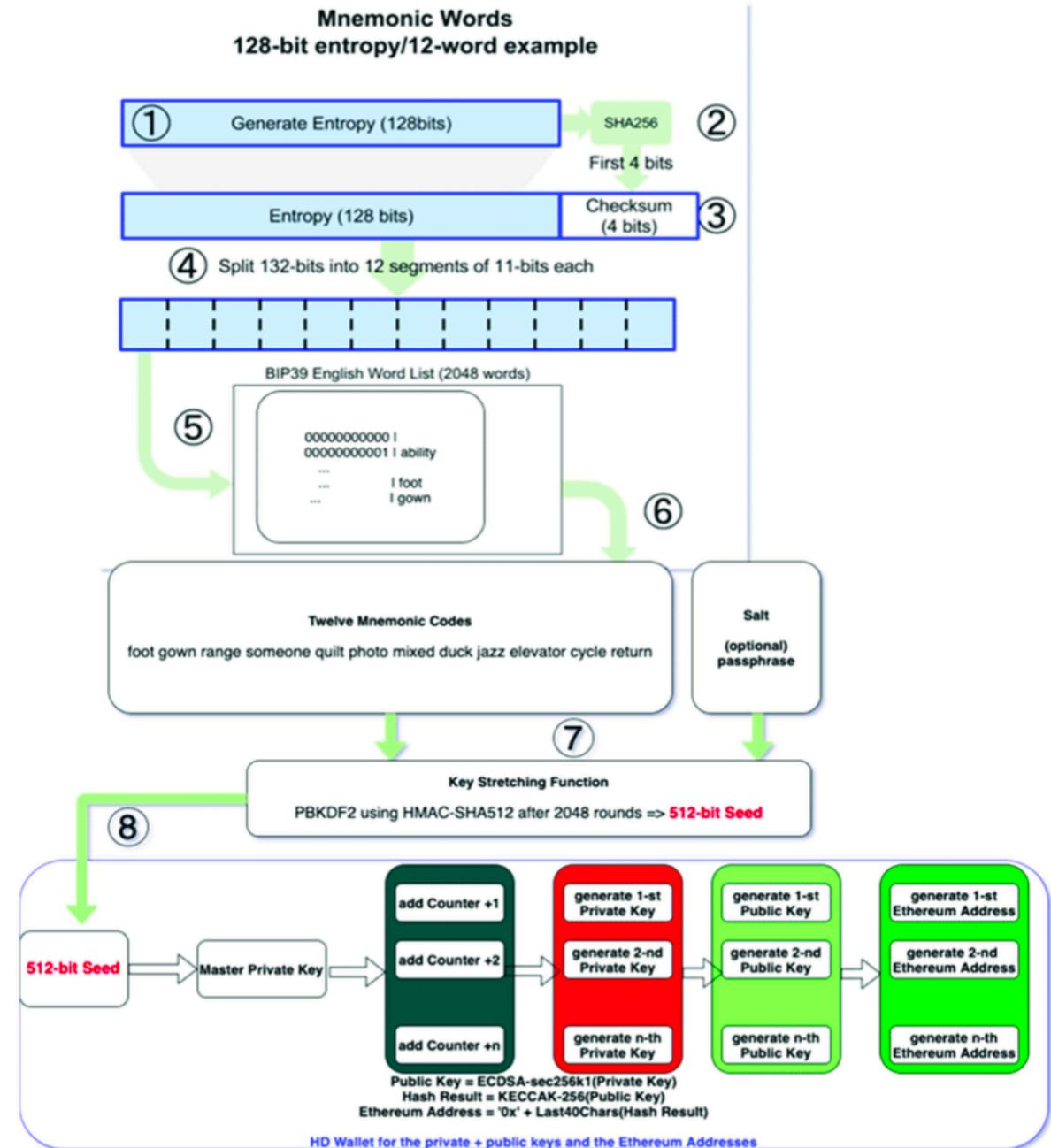
Crypto wallets

What wallets do

- Generate private and public key.
- Store private key.
- Sign and send transactions.
- Show balances.

Mnemonic phrase

A mnemonic phrase is superior for human compared to the handling of raw seed.



Types of wallets

- Cloud
 - Cloud holds private keys.
- Soft
 - Browser extension.
 - Mobile application.
 - Desktop app.
- Hardware
 - Ledger, Trezor, etc.
- Paper
 - Write secret key on paper.
- Brain
 - Memorize your secret key.

Hardware wallets

- Physical devices designed to securely store cryptocurrency private keys offline.
- Connect to a computer via USB and require physical confirmation for transactions.



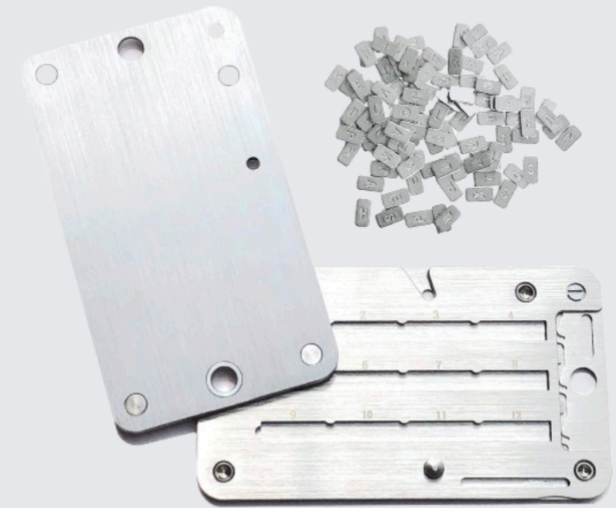
Brain wallet

- Storing private key using a memorized phrase, that was self-generated.
- Have low entropy, vulnerable to brute-force.



Paper wallet

- Write private key on paper (or other physical item).
- Secure from digital attacks, but vulnerable to physical risks like damage, theft, or loss.



Let's create a hot wallet

- **Metamask** is the most popular one.

